P&S HM-4

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1) Ten individual have participated in a diet - modefication to stimulate weigh loss.

Subject →	1	2	3	4	5	6	7	8	9	10
Before →	195	213	247	261	187	210	215	246	294	310
After →	187	195	221	190	175	197	199	221	278	285

Ans Hypothesis

- $H_0 : \mu_d = 0$ (no weight reduction)
- $H_1 : \mu_d > 0$ (weigh reduction).

Difference (Before − After)

$d = [8, 18, 26, 11, 12, 13, 16, 25, 16, 25]$

$$\bar{d} = 17, \quad S_d \approx 6.14, \quad n = 10$$

$$S_d \to \sqrt{\frac{\Sigma(d-\bar{d})^2}{n-1}}$$

Test statistic

$$t = \frac{\bar{d}}{S_d/\sqrt{n}}$$

$$t = \frac{17}{6.14/\sqrt{10}} \approx 8.75 \text{ Ans}$$

Critical $t_{0.05,9} \approx 1.833$

$8.75 > 1.833$

Reject H_0

2. A product developer is intrested - - - -
- - - - Apply appropriate formula for test of hypothesis:-

Ans. Two independent Samples, σ Knows z-test

Hypothesis:-

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

formula.

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\Rightarrow \frac{121 - 112}{8\sqrt{0.2}} = 2.52$$

Critical $z = 1.645$

$2.52 > 1.645$

Reject H_0 → New ingrediant reduces drying time.

3 1000 students at College level were graded
Condition at home.

Economic Condition	High	Low	Total
Rich	460	140	600
Poor	240	160	400
Total	700	300	1000

Ans Concept
Test relationship between two Categorical variables.

* Hypotheses

H_0: No association

H_1: Association exists.

Formula.

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

Expected value.

$$E = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$$

$$\rightarrow \frac{600 \times 700}{1000} = 420$$

$df = 1$ $\chi^2 = 19.84$

4 Confidence interval (Mean)

* Concept

Small Sample → use t-distribution.

Formula

$$\bar{x} \pm t \frac{s}{\sqrt{n}}$$

values

$\bar{x} = 12.04$, $s = 0.34$, $n = 8$

$t_{0.025,7} = 2.365$

Calculation.

$$12.04 \pm 2.365 \times \frac{0.34}{\sqrt{8}} = 12.04 \pm 0.28$$

Final Answer

(11.76, 12.32)

5 There sample obtained form ---- --- are equal.

8	7	12
10	5	9
7	10	13
14	9	12
11	9	14

Ans Hypotheses.

$H_0 : \mu_1 = \mu_2 = \mu_3$

H_1 = At least one mean is different.

Sample Mean:-

$\bar{x}_1 = 10, \bar{x}_2 = 8, \bar{x}_3 = 12$

Grand mean

$\bar{x} = 10$

ANOVA formula

$$F = \frac{\text{Mean Square Between}}{\text{Mean Square within}}$$

→ SSB (Sum of Square)

↳ $n[(\bar{x}_1 - \bar{x})^2 + (\bar{x}_2 - \bar{x})^2 + (\bar{x}_3 - \bar{x})^2]$

→ $5[(10-10)^2 + (8-10)^2 + (12-10)^2] = 40$

→ SSW (Sum of Square within)

→ $30 + 16 + 14 = 60$

→ Degrees of Freedom.
Between: $df_1 = 2$
within: $df_2 = 12$

→ Mean Squares

$MSB = \frac{40}{2} = 20$, $MSW = \frac{60}{12} = 5$

$F\text{-Value} = \frac{20}{5} = 4$

Decision.

Critical value: $F_{0.05}(2, 12) = 3.89$

$4 > 3.89$.

6) Test whether two large sample come from the same population?

Ans Given Data.

$n_1 = 1000$, $\bar{x}_1 = 67.5$

$n_2 = 2000$, $\bar{x}_2 = 68.0$

$\sigma = 2.5$ $\alpha = 0.05$

Hypotheses

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

Test Statistic

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \rightarrow \frac{67.5 - 68.0}{2.5\sqrt{\frac{1}{1000} + \frac{1}{2000}}}$$

$\cancel{\rightarrow 5.}$

$\rightarrow 0.0387$

$\rightarrow -5.17$

Critical value

$\alpha = 0.05$

Z critical $= \pm 1.96$

Decision

$|Z| = 5.17 > 1.96$

II Test difference in petrol mileage of two car types

Given Data

Sm 1

$n_1 = 42, \bar{x}_1 = 15, \sigma_1^2 = 2.0$

Sm 2

$n_2 = 80, \bar{x}_2 = 11.5, \sigma_2^2 = 1.5$

$\alpha = 0.01$

Hypotheses

$H_0 : \mu_1 = \mu_2$

$H_1 : \mu_1 \neq \mu_2$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Substitut values.

$$z = \frac{15 - 11.5}{\sqrt{\frac{2}{42} + \frac{1.5}{80}}}$$

$\rightarrow 1359$

critical value.

$\alpha = 0.01$

$Z_{critical} = \pm 2.58$

$|z| = 13.59 > 2.58$

8. ANOVA

Hypotheses

H_0: All means equal

H_1: At least one mean differers.

Formula. $F = \frac{MSB}{MSW}$

$SSB = 2668.8, SSW = 5939.6$

$df_1 = 2, df_2 = 12$

$MSB = 1334.4, MSW = 494.97$

$F = 2.69$

Decision.

critical value $F_{0.01}(2,12) = 6.93$

$2.69 < 6.93$

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Given: $\mu = 12$
$\sigma = 0.5$
$n = 4$
$\alpha = 0.05$.

Hypotheses

$H_0 : \mu = 12$

$H_1 : \mu < 12$

Test statics.

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

val

$$t = \frac{\bar{x} - 12}{0.25}$$

Critical valu

$df = n - 1 = 3$

$\alpha = 0.05$

$t_{0.05,3} = -2.353$

Decision Rule

$t < -2.353 \rightarrow$ Reject H_0

otherwise $\rightarrow$ Fail to reject H_0

10 Z test for mean.

Given $n = 50$

$\bar{x} = 4.05$

$\mu = 4$

$\sigma = 0.2$

$\alpha = 0.05$

Hypotheses.

$H_0 : \mu = 4$

$H_1 : \mu > 4$

Test Static

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Calculation.

$$Z = \frac{4.05 - 4}{0.2/\sqrt{50}} \approx 1.77.$$

Critical value.

$Z_{0.05} = 1.645$

Decision

$1.77 > 1.645$

Reject H_0.